Personal Information

Yukai Wang

Gender: Male Email: ericwang_yk@outlook.com
Phone: (+86) 18861837371
Target roles: Algorithm Engineer, AI Product Engineer



Education

University of Manchester, Department of Computer Science Manchester, UK
MSc in Artificial Intelligence *2025.9 - 2026.12*
First-class degree

- Data Engineering Technologies
- Topics in Machine Learning
- Transforming Text Into Meaning
- Advanced Topics in Machine Learning

Jiangnan University, School of Artificial Intelligence and Computer Science Wuxi, China
BEng in Artificial Intelligence *2021.9 - 2025.6*
Average grade: 85/100

- Natural Language Processing
- Computer Vision
- Optimisation Theory and Methods
- Machine Learning and Deep Learning
- Computer Graphics
- Data Structures

University of California, Los Angeles, UCLA Extension Los Angeles, USA
Machine Learning Using R (GPA 4.0/4.0) *2024.4 - 2024.6*

Research Experience

University of Manchester, Department of Computer Science Manchester, UK
Supervisor: Prof. Mikel Luján, Arm / Royal Academy of Engineering Research Chair in Computer Systems *2026.6 - Present*
Project: GS-Bench: A Multidimensional Evaluation Framework for Gaussian-Splatting SLAM

- Developed a unified configuration system and algorithm adapters, extending SLAMBench to support Gaussian representations; designed and implemented GS-Bench evaluations covering trajectories, 3D geometry, spatial/temporal rendering and runtime resources.
- Designed a cross-representation evaluation protocol, standardising conversion between Gaussians, point clouds and TSDF/Mesh; combined protocol validation with SHA-256 to ensure result traceability and comparability.
- Proposed Causal CGVQM-SLAM to assess temporal quality using only current and past frames; evaluated 6 SLAM methods on 3 datasets. In the tests, a 16-frame window yielded the smallest mean difference from offline scores.

University of Manchester, Division of Informatics, Imaging and Data Sciences Manchester, UK
Research Assistant Supervisor: Claudia Lindner, Senior Research Fellow and Research Translation Lead, Pankhurst Institute *2025.11 - Present*
Project: Development, Evaluation and Clinical Translation of AI Medical Devices

- Studied the development, evaluation and implementation of medical AI, identifying requirements and barriers in 6 key areas, including data quality and clinical validation.
- Analysed requirements for model generalisability and reliability, focusing on external validation, subgroup bias and model calibration; identified evidence gaps between retrospective performance and real-world clinical benefit.
- Built a cross-stage translational evidence map linking intended use, validation design and post-deployment governance; clarified how early decisions on intended use and risk affect subsequent validation and ongoing monitoring.

Jiangnan University, School of Artificial Intelligence and Computer Science Wuxi, China
Research Assistant Supervisor: Prof. Yinghui Wang *2023.1 - 2024.10*
Project: Wireless Capsule Endoscopy (WCE) for Gastrointestinal Structure Reconstruction and Autonomous Navigation

- Optimised adaptive window sizing based on local variance to improve the consistency of WCE image restoration quality.
- Implemented an image enhancement algorithm for small-intestinal villi.
- Improved IRMLE and PSNR by 12.63 and 45.47, respectively, while reducing NIQE by 31.84.

Jiangnan University, School of Artificial Intelligence and Computer Science Wuxi, China
Software Team Lead Supervisor: Associate Prof. Hengyang Lu *2022.9 - 2024.6*
Project: "Yipaichengwen": Swarm-Intelligence-Based Intelligent SOP Generation System for Non-standard Operating Procedures of Large Instruments
National Undergraduate Innovation and Entrepreneurship Training Programme

- Used artificial intelligence to extract and integrate non-standard operating procedures.
- Applied NLP and deep learning to automatically generate standardised operating specifications; optimised models and algorithms to reduce risk, improve recognition accuracy and enable personalisation.
- Applied similarity fusion based on cosine similarity and voting within ensemble learning.

Internship Experience

National Supercomputing Center in Wuxi Wuxi, China
AI Engineer Intern *2025.3 - 2025.4*
Project: AIGC System Development: Large Language Model Deployment and Applications

- Improved a mixed-precision quantisation scheme and successfully deployed QWQ-32B and DeepSeek-R1-Distill-70B on a GPU with 40 GB of memory.
- Deployed DeepSeek-R1-671B in a multi-GPU environment with 160 GB of total GPU memory using MoE expert offloading and CPU–GPU heterogeneous inference.
- Conducted in-depth research on emerging industry developments, including the feasibility of the KTransformers architecture.

Inspur Group Wuxi, China
Data Governance Engineer Intern *2024.7 - 2024.8*
Project: Urban Government Affairs Large Language Model Development with Baidu AI Cloud

- Preliminary data preparation and processing: screened government database entries to extract question–answer pairs on topical issues.
- Testing, accurate annotation and results evaluation: used an external policy knowledge-base plugin and the Alibaba Tongyi large language model API to generate answers and ensure annotation accuracy.
- Government affairs Q&A model evaluation: compared results with those from Alibaba Tongyi, providing feedback and corrections for different types of errors.

Publications and Outputs

[1] LV-FeatEx: Large Viewpoint-Image Feature Extraction, ***Mathematics***, 2025, Volume 13, Issue 7, 1111, **First author**.

[2] Research on Insurance Policy Formulation Based on ARIMA-KMEANS Algorithm, ***The 2nd International Conference on Financial Mathematics, Investment and Business Management***, 2024, **Second author**.

[3] Capsule Endoscopy Image Enhancement for Small Intestinal Villi Clarity, ***Mathematics***, 2024, Volume 12, Issue 21, 3317, **Co-author**.

[4] Highlight Removal from Wireless Capsule Endoscopy Images, ***IEEE Journal of Translational Engineering in Health and Medicine***, 2025, Volume 13, pp. 365–375, **Co-author**.

[5] "Yipaichengwen": Swarm-Intelligence-Based Intelligent SOP Generation System for Non-standard Operating Procedures of Large Instruments, **Software copyright holder**, 2024, Registration No. 2024SR0961401, **First author**.

Awards and Honours

- Excellence Award, China Education Television's 3rd "Stories of China Retold in English" Challenge (East China Division 1), 2022.
- Jiangnan University Third-Class Scholarships for the 2021–2022 and 2022–2023 academic years.

Skills

- Computing: proficient in C/C++, Python, MATLAB and SQL; familiar with PyTorch, TensorFlow and other machine learning frameworks.
- Languages: English (TOEFL 96).
- Interests: bamboo flute (Grade 8), badminton.

个人信息

王喻恺

性别：男　　邮箱：ericwang_yk@outlook.com
手机：(+86) 18861837371
求职岗位：算法工程师，AI 产品工程师



教育经历

曼彻斯特大学计算机科学系 曼彻斯特，英国
理学硕士——人工智能 *2025.9 - 2026.12*
一等学位
- Data Engineering Technologies（数据工程技术）
- Topics in Machine Learning（机器学习专题）
- Transforming Text Into Meaning（文本语义解析）
- Advanced Topics in Machine Learning（高级机器学习专题）

江南大学人工智能与计算机学院 无锡，中国
工学学士——人工智能 *2021.9 - 2025.6*
平均成绩：85/100
- 自然语言处理
- 计算机视觉
- 最优化理论与方法
- 机器学习与深度学习
- 计算机图形学
- 数据结构

加州大学洛杉矶分校，UCLA Extension 洛杉矶，美国
R 语言机器学习课程（GPA 4.0/4.0） *2024.4 - 2024.6*

科研经历

曼彻斯特大学计算机科学系 曼彻斯特，英国
导师：Mikel Luján 教授，Arm / 英国皇家工程院计算机系统研究讲席教授 *2026.6 - 至今*
项目：GS-Bench：Gaussian-Splatting SLAM 多维评测框架
- 开发了统一配置与算法适配器，升级 SLAMBench 以接入 Gaussian；设计并实现覆盖轨迹、3D 几何、空间/时序渲染及运行资源等 GS-Bench 多维评测。
- 设计跨表示评测协议，规范 Gaussian、点云及 TSDF/Mesh 的转换，结合协议校验与 SHA-256，实现结果追溯与可比性控制。
- 提出 Causal CGVQM-SLAM，仅用当前及历史帧评估时序质量；完成 6 种 SLAM × 3 个数据集实验，16 帧窗口在测试中与离线评分的平均差异最小。

曼彻斯特大学信息学、影像与数据科学部 曼彻斯特，英国
研究助理　导师：Claudia Lindner 高级研究员，Pankhurst 研究所科研转化负责人 *2025.11 - 至今*
项目：医疗 AI 器械开发、评估与临床转化研究
- 研究了医疗 AI 开发、评估与实施，提出数据质量、临床验证等 6 个关键领域的要求与障碍。
- 分析模型泛化与可靠性评估要求，聚焦外部验证、亚组偏差与模型校准，识别回顾性性能与真实临床效益之间的证据缺口。
- 形成跨阶段转化证据图谱，关联预期用途、验证设计与部署后治理，明确早期用途与风险决策对后续验证及持续监测的影响。

江南大学人工智能与计算机学院 无锡，中国
研究助理　导师：王映辉教授 *2023.1 - 2024.10*
项目：无线胶囊内窥镜（WCE）在胃肠道结构恢复与自主导航中的应用
- 优化局部方差自适应窗口尺寸，提高了 WCE 图像恢复质量的均匀性。
- 实现了小肠绒毛图像增强算法。
- IRMLE 和 PSNR 分别提高 12.63 和 45.47，同时将 NIQE 降低 31.84。

江南大学人工智能与计算机学院 无锡，中国
软件团队负责人　导师：陆恒杨副教授 *2022.9 - 2024.6*
项目："仪拍成文"——基于群体智能的大型仪器非标准操作流程 SOP 智能生成系统
国家级大学生创新创业训练计划
- 利用人工智能技术对非标准化操作流程进行提取和整合。
- 应用自然语言处理深度学习技术实现标准化操作规范的自动生成，并优化模型和算法以降低风险、提高识别准确率并实现个性化定制。
- 在集成学习中应用基于余弦相似度和投票法的相似性融合算法。

实习经历

无锡国家超级计算中心 无锡，中国
人工智能工程师实习生 *2025.3 - 2025.4*
项目：AIGC 系统构建——大型语言模型部署与应用
- 改进了混合精度量化方案，成功在 40GB GPU 上部署了 QWQ-32B 和 DeepSeek-R1-Distill-70B 模型。

- 采用 MoE 专家卸载与 CPU–GPU 异构推理，在总显存 160 GB 的多 GPU 环境中部署了 DeepSeek-R1-671B 模型。
- 对行业前沿发展进行深入研究，包括 KTransformers 架构的可行性研究。

浪潮集团 无锡，中国
数据治理工程师实习生 *2024.7 - 2024.8*
项目：与百度智能云合作的城市政务大模型构建

- 初步数据准备与处理：从政府数据库中筛选条目，提取热点话题的问答对。
- 测试、精准标注与结果评估：应用外部政策知识库插件和阿里通义大模型接口生成答案，确保标注的正确性。
- 政务问答模型评估：通过与阿里通义大模型的结果对比进行评估，并针对不同类型的错误提供反馈和修正。

发表成果

[1] LV-FeatEx: Large Viewpoint-Image Feature Extraction, ***Mathematics***, 2025, Volume 13, Issue 7, 1111, **第一作者**.

[2] Research on Insurance Policy Formulation Based on ARIMA-KMEANS Algorithm, ***The 2nd International Conference on Financial Mathematics, Investment and Business Management***, 2024, **第二作者**.

[3] Capsule Endoscopy Image Enhancement for Small Intestinal Villi Clarity, ***Mathematics***, 2024, Volume 12, Issue 21, 3317, **合作者**.

[4] Highlight Removal from Wireless Capsule Endoscopy Images, ***IEEE Journal of Translational Engineering in Health and Medicine***, 2025, Volume 13, pp. 365–375, **合作者**.

[5] 《“仪拍成文”——基于群体智能的大型仪器非标准操作流程 SOP 智能生成系统》，**软件著作权所有者**，2024 年，登记号 2024SR0961401，**第一作者**。

荣誉奖项

- 2022 年中国教育电视台第三届“用英语讲中国故事”挑战赛（华东一区）优秀奖。
- 江南大学 2021-2022 学年、2022-2023 学年三等奖学金。

技能能力

- 计算机：精通 C/C++、Python、MATLAB、SQL 编程语言；熟悉 PyTorch、TensorFlow 等机器学习框架。
- 语言：英语（托福 96 分）。
- 爱好：竹笛（8级）、羽毛球。